
Effect of Memory Support and Elicited Production on Fast Mapping of New Words by Adolescents With Down Syndrome

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Purpose: The purpose of this study was to determine whether memory support and elicited production differentially benefited fast mapping of new vocabulary (comprehension, production accuracy, and speed) in adolescents with Down syndrome (DS) compared with typically developing (TD) children matched for syntax comprehension. The study also examined predictors of individual difference within groups.

Method: The ability of 19 adolescents with DS to fast map new noun vocabulary was compared with 18 TD children matched for syntax comprehension. The effects of memory support (5 examiner repetitions of the novel word vs. 1) and elicited production (2 repetitions by child vs. none) were evaluated in a repeated-measures design with counterbalancing of tasks.

Results: For novel word comprehension, low memory support impaired the TD but not DS participants. DS participants were slower to respond correctly in all conditions but benefited more from memory support than TD participants in speed of response. For novel word production, high memory support benefited all participants in the no-elicited-production condition. Elicited production improved performance equally in the 2 memory conditions. Standard comprehension measures predicted fast-mapped comprehension: vocabulary in the case of the DS group, syntax in the case of the TD group. Auditory short-term memory measures predicted fast-mapped production in the TD group but not the DS group. Hearing and grammatical morpheme comprehension predicted novel word production in the DS group.

Conclusions: DS participants' speed of comprehension of fast-mapped words was differentially increased compared with the TD syntax-comprehension-matched group, by memory support. Elicited production improved production fast-mapping for all participants, and memory support improved it in the absence of elicited production. Comprehension predicted individual differences in fast-mapped comprehension in DS; in addition, hearing predicted production.

KEY WORDS: fast mapping, Down syndrome, auditory short-term memory

In this study we investigated the fast-mapping skills of adolescents with Down syndrome (DS) compared with typically developing (TD) children matched for syntactic comprehension. We ask whether the increased gap between syntax comprehension and production that occurs for individuals with DS also occurs for learning to comprehend and produce new vocabulary and whether increased opportunities to hear the novel words (memory support) or say them (elicited production) differentially improve accuracy and speed of recall for the group with DS.

Specific Behavior Phenotype for Language and Cognition in DS

Individuals with DS typically have three copies of Chromosome 21. As a consequence of 21 trisomy and the effects of differential gene expression on fetal development and physiological function (Hattori et al., 2000), persons with DS show both wide individual variation in cognitive and communicative development and a phenotypically specific pattern of within-individual strengths and weaknesses (Chapman & Hesketh, 2000; Dykens, Hodapp, & Evans, 1994; Eadie, Fey, Douglas, & Parsons, 2002; Miller, 1995; Rosin, Swift, Bless, & Vetter, 1988). The phenotype differs from the patterns of individuals with other syndromes of cognitive disability, including those with fragile X (Abbeduto et al., 2001) and Williams syndromes (Bellugi, Lichtenberger, Jones, Lai, & St. George, 2000). The pattern in DS is one of cognitive disability, with additional deficits in auditory working memory and in expressive syntax and lexicon compared with comprehension of syntax and vocabulary. Syntax is more affected than vocabulary, and syntax comprehension skills decline relative to nonverbal cognition in adolescence. For this reason, we selected a comparison group of TD children matched on the basis of syntax comprehension, which allowed us to ask whether the comprehension–production gap in syntax would also be evident in new vocabulary learning for adolescents with DS compared with the TD group. We selected syntax comprehension rather than vocabulary comprehension as a matching variable because the latter tests can yield widely different levels of performance, depending on their construction (Miolo, Chapman, & Sindberg, 2005) and because fast mapping requires sentence comprehension as well as lexical comprehension to identify the novel word part of speech.

Fast Mapping

The fast-mapping task (Carey & Bartlett, 1978; Dollaghan, 1985, 1987; Rice, 1990; Wilkinson, Dube, & McIlvane, 1996) includes one or more mentions of a novel word for a novel object in the context of another task (e.g., hiding objects) in which learning is incidental. The fast-mapping paradigm has been extended to multiple presentations of the novel word in incidental learning contexts (*quick incidental learning*) by Rice and colleagues (Rice, 1990; Rice, Buhr, & Nemeth, 1990).

Evidence for a fast-mapping deficit is mixed. In earlier research, children and adolescents with DS have shown the ability to fast map a single CVC novel word in comprehension and production at levels commensurate with their nonverbal mental age in a hiding task (Chapman, Kay-Raining Bird, & Schwartz, 1990). Young

children with DS were able to extend the novel term to other examples of a novel category (Mervis & Bertrand, 1995). These successes did not require additional memory support or elicited production, despite the evidence of an expressive language deficit in word use, syntax, and grammatical morphology (see Chapman, 2003; Chapman & Hesketh, 2000; Roizen, 2001, for reviews; Eadie et al., 2002).

When multiple items must be successively fast mapped (e.g., six hiding tasks in succession), however, the group with DS has more difficulty than nonverbal mental age and comprehension control participants on production, but not comprehension, testing (Chapman, 2003). We predicted that our results in this study would provide evidence of a production deficit for vocabulary fast mapping when the gap between comprehension and production was compared with performance in TD children on multiple items.

Task variables. Perceptual representation of the novel word requires relatively little detail in the fast-mapping paradigm because the comprehension task is typically a recognition one and may require as little knowledge as the fact that this was both a novel and a named object. Kay-Raining Bird and Chapman (1998) found, by testing novel words of decreasing similarity to the target word, that the fast-mapped phonological representation of toddlers was incomplete, but the test paradigm might make those difficulties apparent only in production, rather than comprehension. In this study, we added an additional control for comprehension testing in which two of the novel objects had both been named in the task; thus, the comprehension task could not be solved simply by memory for which novel object has been labeled. A third novel object, not previously named or encountered, was added at test time, to ensure that sheer object novelty was not the basis for choice in the comprehension task. The hiding task was framed as one of practicing magic tricks, to make it pragmatically appropriate for all participants.

Experimental Variables

Memory support. Auditory short-term memory is also an area of selective deficit in individuals with DS (Eadie et al., 2002; Jarrold & Baddeley, 1997, 2001; Jarrold, Baddeley, & Hewes, 1999; Jarrold, Baddeley, & Phillips, 2002; Kay-Raining Bird & Chapman, 1994; Kittler, Krinsky-McHale, & Devenny, 2004; Lanfranchi, Cornoldi, & Vianello, 2004; Laws, 1998; Marcell & Weeks, 1988). One might expect that task manipulations that increase the strength of short-term representations in phonological store (i.e., memory support) would improve performance differentially for the DS group compared with TD children matched for syntax comprehension.

As many as four presentations are sufficient for comprehension fast mapping of 13- to 16-month old infants who are TD; an increase to eight presentations does not improve performance (Kay-Raining Bird & Chapman, 1998). Multiple presentations of words (*modeling*) have also been shown to be an effective language intervention method for some individuals with specific language impairment (Weismer & Murray-Branch, 1989; Weismer, Murray-Branch, & Miller, 1993). In this study, we examined the effects of memory support (the examiner's repetition of a novel word five times vs. one time) on the ability of adolescents with DS to fast map new vocabulary. In addition, we examined effects on naming latency. Digit-span repetition latencies are typically slower in DS individuals (Seung & Chapman, 2000) but confrontation naming latencies for familiar objects are not (Marcell, Busby, Mansker, & Whelan, 1998). Whether latencies are slower for fast-mapped novel words, with their less automatized representation in phonological store, is not known.

Elicited production. Children's spontaneous imitations of new vocabulary have been associated with subsequent learning in both TD (Leonard, Schwartz, Folger, Newhoff, & Wilcox, 1979) and language-impaired children (Schwartz & Leonard, 1985). Asking children to imitate (*elicited* or *evoked production*) has been used effectively in language intervention, although embedding opportunities for production in natural conversation has proven even more effective (Camarata, Nelson, & Camarata, 1994). Case studies have indicated that combining evoked production with modeling of words succeeded for some individuals with specific language impairment (Weismer & Murray-Branch, 1989; Weismer, Murray-Branch, & Miller, 1993). In teaching object labels to preschoolers with mental retardation, elicited production used under conditions of following the child's lead is more successful than when the child's attention to the object is recruited (Yoder, Kaiser, Alpert, & Fischer, 1993). In this study, we contrasted no requests for elicited production with two requested repetitions. If the auditory-verbal short-term memory (AVSTM) deficit is linked to poorer articulatory representation (or functioning of the articulatory loop, following Baddeley's, 2000, model), we would expect individuals with DS to benefit differentially from elicited production compared with TD children matched for syntax comprehension in naming accuracy and speed.

Predicting Individual Variability

Hearing. Recent work in the fast-mapping paradigm has also demonstrated a role for hearing in the early stages of vocabulary learning. Toddlers (26–28 months) with otitis media with effusion (OME) have particular difficulty in learning to discriminate voiced–

voiceless *z*–*s* pairs, in fast mapping (Petinou, Schwartz, Gravel, & Raphael, 2001). Because individuals with DS often have difficulty with OME or sensorineural hearing loss (estimates range from 40% to 80%, mostly OME related, according to Shott, Joseph, & Heithaus, 2001), they might be expected to have more difficulty in developing perceptual representations of words, particularly those encountered only once, than comprehension-matched control participants. Variability in accuracy of comprehension, and latency of correct production, might be related to hearing status in the DS group.

Phonological working memory. Phonological working-memory measures been shown to play a predictive role in the syntactic language skills of individuals with DS (Chapman, Hesketh, & Kistler, 2002; Miolo et al., 2005) and in the fast mapping of production vocabulary (Chapman, 2003). In this study we investigated whether experimental variations that might support auditory short-term memory in comprehension or production could improve the fast mapping of new vocabulary and whether the split between comprehension and production skills shown in syntax for individuals with DS could also be found in fast mapping of vocabulary.

Recent models of working memory argue for multiple components that might be responsible for the observed relationship. In Baddeley's (2000) model, phonological working memory includes phonological store, the articulatory rehearsal loop, central executive skills, and (added recently) episodic memory recruiting long-term linguistic knowledge, particularly phonotactics. (A visual scratch-pad constitutes the other part of working memory.) In Cowan et al.'s (2003) recent work, working-memory performance was parsed into three components: a general working-memory component independent of general intelligence and related to span, knowledge of the particular domain (which relates to general intelligence and might be equated with Baddeley's episodic store), and processing speed, which in Baddeley's model is assumed to play a role in how much material can be rehearsed in the articulatory loop. Latency of recall is one such index of processing speed (Cowan et al., 2003).

In regression analyses in this study, we evaluated the relation of auditory-memory span and hearing, which we expected to relate, through phonological store, to comprehension speed and accuracy. We anticipated that individual variability in fast-mapped, novel word production would be predicted for the DS group, by AVSTM span, as indexed by a digit-span task. We expected novel word comprehension, in contrast, to be predicted by syntax comprehension, on the basis of prior research with individuals with DS who were in the 5- to 27-year-old age range (Chapman, 2003).

This study compared adolescents with DS and TD control participants who were matched for syntax comprehension on fast-mapping skill. We asked (a) whether memory support (five vs. one novel word presentations) and elicited production (two vs. zero) would benefit adolescents with DS disproportionately compared with control participants in comprehension, production accuracy, and latency and (b) whether the gap between comprehension and production accuracy would be greater for the group with DS. Hearing status and AVSTM span were evaluated as predictors of individual differences in performance in each group.

Method

Participants

Participants included 19 adolescents and young adults with DS, ages 12 through 21 years, and 18 children, ages 2 through 5 years, who were TD and matched to the DS group on syntax comprehension (assessed by mean of age-equivalent scores on Test for Auditory Comprehension of Language—Third Edition [TACL-3; Carrow-Woolfolk, 1999] Subtests II [Grammatical Morphemes] and III [Elaborated Phrases]). Participant characteristics are summarized in Table 1. Participants were recruited throughout Wisconsin and northern Illinois through the Human Assessment core at the Waisman Research Center and reimbursed for their time at the rate of \$10 per hr.

Participants were excluded if spoken English was not the principle means of communication, if a diagnosis other than DS had been assigned (e.g., fragile X, cerebral palsy, autism, fetal alcohol syndrome, psychiatric disorder, attention deficit disorder), if any hearing loss was greater than a 40-dB threshold pure tone average (PTA) unilaterally, if the participant could not engage in activities that lasted up to 20 min, or if mean length of utterance in the narrative transcript was less than 2.0.

Procedures

During the first 3-hr session of seven sessions, a series of standardized tests of nonverbal cognition, language comprehension, and working memory were administered, together with a hearing screening and the protocol for the experimental fast-mapping task described below.

Hearing. Hearing was screened bilaterally at 20 dB under headphones by a licensed audiologist. Screening was performed at 500, 1000, 2000, and 4000 Hz. If a participant did not respond at 20 dB at any level, threshold testing was conducted for that level. Participants were included in the study only if any hearing loss was less than 40-dB PTA unilaterally. Tympanometric testing was also performed and the hearing assessment results shared immediately with the family.

Standardized tests. Visual nonverbal cognition was assessed by the Pattern Analysis Subtest of the Stanford–Binet Intelligence Scale, Fourth Edition

Table 1. Participant characteristics: means (standard deviations).

Variable	Participants with DS	Syntax-comprehension-matched	Cohen's effect size (d)	r
Number	19	18		
% girls	47	61		
Chronological age (years)	15.94 (2.08)	4.00 (0.76)		
Mother's years of education	15.32 (2.19)	16.89 (1.97)		
% who passed hearing screen	47	89		
Syntax comprehension ^a	5.07 (1.11)	5.01 (1.08)	0.05	.03
Vocabulary comprehension ^b	5.23 (1.47)	5.46 (1.64)	−0.15	−.07
Nonverbal visual cognition ^c	5.13 (1.40)	3.82 (0.59)	1.22	.52
Visual short-term memory ^d	4.71 (1.20)	4.36 (1.13)	0.30	.15
Auditory-verbal short-term memory ^e	4.47 (1.67)	4.42 (1.30)	−0.03	.02
NRT (phonemes)	56.16 (13.71)	64.72 (16.34)	−0.57	.27

Note. *n* = 15, typically developing group. NRT = Nonword Repetition Task (Dollaghan & Campbell, 1998).

^aTACL-3 (Test for Auditory Comprehension of Language—Third Edition [Carrow-Woolfolk, 1999]) = mean of age-equivalent scores on Subtest II, Grammatical Morphemes, and Subtest III, Elaborated Phrases. ^bTACL-3 = age-equivalent score on Subtest I, Vocabulary Comprehension. ^cAge-equivalent score on Pattern Analysis subtest of the Stanford–Binet Intelligence Scale (4th ed. [SB4]; Thorndike, Hagen, & Sattler, 1986). ^dAge-equivalent score on Bead Memory subtest of the SB4. ^eAge-equivalent score on the Digit Span subtest of the Kaufman Assessment Battery for Children (Kaufman & Kaufman, 1983).

(Thorndike, Hagen, & Sattler, 1986), which requires matching of abstract patterns in blocks. Visual working memory was assessed by the Scale's Bead Memory subtest, which requires reconstruction of a bead sequence after the model is removed. Language comprehension was assessed by the TACL-3 (Carrow-Woolfolk, 1999), which requires the participant to point to the correct picture out of three. Subtest II, Grammatical Morphemes, and Subtest III, Elaborated Phrases, were used to establish syntax comprehension, and Subtest I, Vocabulary, was used to assess vocabulary comprehension. The Peabody Picture Vocabulary Test—III (PPVT—III; Dunn & Dunn, 1997) was also administered for comparison (see Miolo et al., 2005). Phonological working memory was assessed by the Digit Span subtest of the Kaufman Assessment Battery for Children (K-ABC; Kaufman & Kaufman, 1983), which is thought to reflect short-term phonological store and articulatory rehearsal processes, and the Nonword Repetition Test (NRT; Dollaghan & Campbell, 1998), which is thought to reflect both short-term phonological store and the contribution of long-term phonotactic learning. The role of these measures in predicting sentence comprehension is reported elsewhere (Miolo et al., 2005). When administration procedures recommended starting or ending test points on the basis of chronological age, the points were modified, for individuals with DS, to start at the test's beginning and to test until a ceiling sequence of items had been failed (e.g., three items in a row on the K-ABC). Three TD participants were mistakenly not tested to K-ABC ceiling; their data are omitted here. Testing was also ended after one or more failures if the individual began to show extreme frustration.

Order of administration. These and other tasks were administered by two examiners in a 3-hr session, beginning with hearing screening and alternating listening and speaking tasks. The order of comprehension and digit span tasks was counterbalanced across participants. The fast-mapping tasks described below were administered midway through the session. Snack breaks were incorporated.

Fast-Mapping Tasks

Four fast-mapping tasks were presented, each incorporating two familiar and two novel CVC words, for a total of eight novel words. The participant and examiner sat at 90° to each other at a table on which four hiding places and four or five objects could be displayed. Each task consisted of an exposure trial for the two novel words that varied (a) the amount of novel word repetition by the examiner (high vs. low memory support) and (b) whether the participant was asked to repeat the word (elicited production vs. none): One of these conditions was held constant while the other

varied for each task. A hiding trial followed, in which the two familiar objects and the two novel objects were each concealed under one of the four hiding places (a top hat, scarf, bag, and barrel) in a spatial order that varied for each task, with counterbalancing across the tasks. Immediately after, ability to produce the familiar and novel words, comprehend the familiar and novel words, and remember the hiding location of each object was tested. The entire exposure procedure took about 25 min to present. Delayed testing repeated the three tests for each word and object after a filled interval of about 1 hr in the protocol.

Conditions. High memory support consisted of four repetitions of the novel word by the examiner during the exposure trial. Low memory support consisted of a single presentation of the novel word during the exposure trial. Elicited production consisted of two requests to repeat the name of the novel object during the exposure trial, or no elicited-production requests were made.

Tasks. Task 1 varied memory support for the two words (high vs. low), without requiring elicited production. Task 2 varied memory support (high vs. low) and required elicited production of each. Task 3 varied requirements for elicited production (yes, no) and consistently provided high memory support. Task 4 varied requirements for elicited production (yes, no) and consistently provided low memory support. Order of tasks was counterbalanced across participants.

Novel words. Novel words were constructed from early-acquired consonants and vowels in a CVC format; the set used was /kob/, /zəg/, /ted/, /wIs/, /pon/, /flk/, /muk/, and /vaut/. Four different counterbalanced assignments of these words to task conditions were constructed and assigned, sequentially, to participants tested in group. Thus, all novel words occurred, across participants, approximately equally in each task condition.

Novel objects. The novel objects used in each task included pieces of toys with protrusions of various shapes; a handle with gears; a circular box with a pattern of holes in the top; a collapsible wire sphere; a plate with a handle for turning and pieces that moved up and down, or back and forth; linked ball and socket pieces; abstract patterns created with interlocking blocks; a curved, plastic orange peeler; a zest grater; and a stick with a ring over it. Object assignment was counterbalanced to tasks and to foil or named object.

Familiar words and objects. Two familiar objects, and their labels, were used in each task: for example, a watch and coin, string and (playing) card; candle and dice; and bird and rabbit.

Exposure instructions. The tasks were presented as examples of a magician friend's hiding tricks, to put them in a pragmatically appropriate context for the teenagers. A photograph of the magician in a cape and

hat and holding a wand accompanied this explanation, followed by the exposure trial presenting both familiar and novel objects. The familiar objects were labeled as *this*. High-memory-support instructions for the novel objects included four mentions of the novel word in four different carrier phrases: for example, “In his magic shows the magician always hides the /pon/. Here’s the /pon/. He hides the /pon/ behind one of these [sweep hand over hiding places]. I’m glad we have this /pon/.” Low-memory-support instructions mentioned the novel word a single time: for example, “In his magic tricks he also uses the /flk/.” Elicited-production requests, when part of the exposure trial, occurred following this introduction of objects with, for example, “What was this called? [child answers]. Can you tell me the name again?” The responses to these requests were recorded in broad IPA transcription.

Hiding trial. The four hiding locations were then randomized and the participant asked to hide the (word) in the (hiding location). Order of mention was counter-balanced across task orders.

Immediate production test. Participants were then asked, “Let’s check that everything is still here. What is under here?”, and the hiding places were lifted to show each object in turn. The child’s answer was recorded online in broad IPA. Latency of response was recorded from the time-stamped videotape.

Immediate comprehension test. A third novel object, not previously seen or labeled, was then added to the set of objects, for a total of five, and the objects were rearranged. Participants were told, “Now it’s your turn to show me the different things that my magician friend uses in his magic tricks. Give me the [object],” where the objects named were the familiar and novel objects used in the exposure task. The objects selected in response were recorded. Latency for picking up the novel objects was scored from the time-stamped videotapes.

Immediate location test. The four objects were then returned to the table and the participant asked, as the examiner pointed to each object, “Where did you hide this before?”

Delayed testing. The delayed testing was introduced at the end of the 1st day’s testing session, approximately an hour after the immediate fast-mapping tasks, and the sequence of testing was repeated for each of the four tasks.

Delayed production testing. The instruction was, “Remember that I showed you some of the things that my magician friend uses when he does magic tricks? I want you to look at them again to make sure that we have everything.” The four objects were then spread out and the child asked, “Let’s check that everything is still here. What is this?”, with each object indicated in turn. The child’s label was recorded online in broad IPA and

the latency of production scored from the time-stamped videotape.

Delayed comprehension testing. Delayed comprehension testing followed. A third novel object was added, the spatial order of objects rearranged, and the child asked, “Now it’s your turn to show me the different things that my magician friend uses in his magic tricks. Give me the [X],” with the four objects named in turn. The choices for the familiar and fast-mapped words were recorded online and the latency of novel object choice scored from the time-stamped videotape.

Delayed memory for location was then questioned by replacing the four hiding locations on the table and asking, for each of the four objects in turn, “Where did you hide this before?”

Transcription of Responses

Children’s labels for objects were transcribed online using broad IPA and retranscribed from the audiotape, with a check against the online version. (Videotape was also used, if necessary, to see the child’s production of the word.) The objects chosen in the comprehension task and the locations chosen in the location task were recorded online on the score sheet.

Correct responses. Novel word production was scored as correct when two thirds of the phonemes were correct and in the correct order or when all were correct in order and only one additional phoneme was added. Comprehension responses were scored as correct when the object corresponding to the word was indicated. Hiding location responses were scored as correct when the original hiding place for the object was indicated.

Latency of response. Latencies of correct and incorrect responses were measured by two independent coders from the time-stamped videotape for production and comprehension responses. The mean of the two measures was scored as the latency of response time. The time stamp indicated seconds and tenths of a second. The latency of a production response was measured from the end of the examiner’s final inquiry to the beginning of the participant’s production of the target word or the beginning of an *I don’t know* response or gesture. The latency of a comprehension response was measured from the end of the examiner’s inquiry to the end of the participant’s indicating gesture (full extension of pointing response, contact with object chosen, or the full extension of the arm, if the object was being held prior to the question).

Reliability of online IPA transcription. Reliability of IPA transcription was determined by a second transcriber who independently transcribed the audiotapes of the session; agreement with online transcription in phonemes transcribed averaged 87% for both the DS

sample and the TD sample. Intercode agreement in scoring of IPA transcriptions as correct (i.e., two out of three phonemes correct, in order, or addition of only one phoneme) averaged 95% for the DS sample and 96% for the TD sample.

Reliability of latency scoring. For a 15% sample of the latency data, the two coders rescored the latency of response from the time-stamped videotape and means were recomputed. The two means agreed with each other within 0.2 s for 74% of the samples, within 0.5 s for 77% of the samples, and within 1.0 s for 83% of the samples.

Results

Effects of Group, Memory Support, and Elicited Production

The immediate measures of correct production, comprehension, and location of the novel words or objects were analyzed in mixed analyses of variance (ANOVA) designs: Group (DS or TD) $\times$ Memory Support (high or low) $\times$ Elicited Production (some or none), with repeated measures on the last two factors. Alpha was set at .05. These analyses were also carried out on the delayed measures of correct production, comprehension, and location for the participants who completed

the delayed measures (2 DS and 3 TD participants did not complete the delayed tasks, because of the length of other protocol measures). Performance on these measures is summarized by group and condition in Table 2.

Fast mapping in comprehension. For immediate comprehension, there was a significant effect of the interaction of memory support and group, $F(1, 35) = 9.52, p < .01$, partial $\eta^2 = .21$. The means are shown in Table 2. High memory support helped the TD children but not the participants with DS. For delayed comprehension, no interactions or main effects were significant. These analyses provide no evidence that elicited production helped comprehension.

Fast mapping in production. For immediate production, there was a significant main effect of memory support, $F(1, 35) = 6.22, p < .02$, partial $\eta^2 = .15$, and a significant interaction of memory support and elicited production, $F(1, 35) = 4.36, p < .05$, partial $\eta^2 = .11$. The interaction is shown in Table 2. Participants benefited from high memory support when no elicited production was required (.43 vs. .05 out of two items) but not when elicited production was requested (.30 vs. .27). The effect of group was not significant. For delayed production, recall averaged only 5% for the group with DS and 2.5% for the TD group (see Table 2); no ANOVA was conducted.

Table 2. Means (standard deviations) of experimental conditions by group (two maximum each).

Condition	Down syndrome ^a (<i>n</i> = 19)	Syntax-comprehension-matched ^b (<i>n</i> = 18)
Immediate comprehension (8 max.)	4.95 (2.01)	4.56 (1.85)
High memory–no elicited production	1.11 (0.74)	1.33 (0.69)
High memory–elicited production	1.16 (0.69)	1.44 (0.62)
Low memory–no elicited production	1.32 (0.75)	0.83 (0.79)
Low memory–elicited production	1.37 (0.68)	0.94 (0.73)
Immediate production (8 max.)	1.53 (1.17)	0.83 (0.99)
High memory–no elicited production	0.47 (0.61)	0.39 (0.70)
High memory–elicited production	0.37 (0.60)	0.22 (0.43)
Low memory–no elicited production	0.11 (0.32)	0.00 (0.00)
Low memory–elicited production	0.42 (0.51)	0.11 (0.32)
Delayed comprehension (8 max.)	4.12 (2.47)	3.80 (1.52)
High memory–no elicited production	1.12 (0.86)	0.87 (0.74)
High memory–elicited production	1.18 (0.64)	0.93 (0.59)
Low memory–no elicited production	0.94 (0.75)	0.93 (0.88)
Low memory–elicited production	0.88 (0.86)	1.07 (0.70)
Delayed production (8 max.)	0.41 (1.00)	0.20 (0.56)
High memory–no elicited production	0.06 (0.24)	0.13 (0.35)
High memory–elicited production	0.12 (0.33)	0.07 (0.26)
Low memory–no elicited production	0.06 (0.24)	0.00 (0.00)
Low memory–elicited production	0.18 (0.39)	0.00 (0.00)

^a*n* = 17 for delayed comprehension and production. ^b*n* = 15 for delayed measures.

Examination of performance by hearing status in the DS group revealed that those who failed the hearing screen benefited from elicited production in the low-memory condition, but those who passed did not benefit.

The gap between comprehension and production. The two groups were compared, in a one-way ANOVA, on the difference between immediate fast-mapped comprehension and production scores, to determine whether the greater DS gap between comprehension and production of syntax was replicated for fast mapping of vocabulary. The difference between groups was not significant ($p > .01$); the average difference for all participants between the two tasks was 3.60 ($SD = 0.29$) and it did not correlate with the AVSTM measures in either group.

Latency Measures

The few extremely long (8–20 s) latencies were trimmed to 6 s, a value two standard deviations above the mean of the cells with longest latency. Mean latency measures were analyzed for successful word comprehension and production in mixed ANOVA designs: Group (DS or TD) $\times$ Memory Support (high or low) $\times$ Elicited Production (some or none), with repeated measures on the last two factors. Alpha was set at .05.

Latency of comprehension responses. A significant main effect of group, $F(1, 16) = 12.02$, $p < .003$, $\eta^2 = .43$, and a significant interaction of Memory $\times$ Group, $F(1, 16) = 4.87$, $p < .04$, $\eta^2 = .23$, were found. The group with DS was slower to respond in all conditions than the TD group. Memory support benefited the speed of response for the DS group but not the TD group; mean response time was 1.44 s for participants with DS in the high-memory-support condition and 2.02 s in the low-memory-support condition. For the TD participants, means were 1.07 s and 0.91 s, respectively.

Mean latency of correct comprehension response to novel words was also compared with mean latency for the familiar words used as foils in a two-way, mixed-design ANOVA: Group (DS vs. TD) $\times$ Familiarity (novel vs. common words), with repeated measures on the latter factor. Alpha was set at .05. A main effect was found for familiarity, $F(1, 33) = 25.41$, $p < .001$, partial $\eta^2 = .44$. Mean response time to common nouns was 0.85 s ($SD = 0.41$), and to novel words, 1.40 s ($SD = 0.64$). The interaction of familiarity and group was also significant, $F(1, 33) = 5.44$, $p < .03$, partial $\eta^2 = .14$. The group with DS was as fast as the TD group in demonstrating comprehension of common nouns (0.82 s vs. 0.88 s) but significantly slower in responding successfully to novel words (1.64 s vs. 1.18 s).

Latency in production. Participants giving correct responses within conditions were relatively few, mak-

ing a repeated-measures analysis of latency of correct productions impossible.

The group with DS did not differ significantly ($p < .05$) from the TD group in latency of correct production of common nouns: 1.04 s ($SD = 0.78$) versus 0.72 s ($SD = 0.35$).

Predictors of Fast-Mapping Performance

Because we had prior evidence indicating that syntax comprehension would play a role in fast-mapped comprehension and AVSTM measures would play a role in fast-mapped production, we set alpha at .05, one tailed, in the following analyses.

Fast-mapped comprehension. We had anticipated, from prior studies (Chapman, 2003), that syntax comprehension scores would predict fast-mapping comprehension and that auditory short-term-memory scores would predict fast-mapping production of words. In the DS group, total fast-mapped comprehension scores were correlated significantly ($p < .05$, one tailed; Pearson product-moment $r = .42$) with Vocabulary raw scores from the TACL-3 but not with Subtests II or III of the TACL-3, which assess syntax comprehension. The correlation with raw-score PPVT-III was also significant ($r = .46$). Other variables (i.e., chronological age, hearing status, NRT, K-ABC, Stanford-Binet pattern analysis, and Bead Memory) were not significantly related.

In the TD group, raw-score TACL-3 Subtests II and III (Grammatical Morphemes and Elaboration of Sentences) were significantly related ($p < .05$, one tailed) to total fast-mapped comprehension ($r = .46$ and $r = .44$, respectively), with other variables not related. Thus, comprehension performance on standardized tests predicted comprehension in the fast-mapping task, although the relationship appeared to be significant only for Vocabulary in the DS group and for syntax in the TD group.

Fast-mapped production. Our expectation that AVSTM span would predict fast-mapped production was only partially correct. As expected, the DS group performed more poorly than the TD group ($p < .05$, one tailed) on total NRT score, a measure of participants' ability to repeat novel words. Univariate ANOVAs comparing group scores at each of the syllable lengths from one to four revealed significant ($p < .05$, one tailed) differences at both one- and four-syllable lengths. In the DS group, total fast-mapped production scores were not correlated significantly, however, with the NRT ($n = 19$) or with K-ABC Digit Span ($p < .05$, one tailed, $n = 17$). Instead, hearing status ($r = .39$) and the TACL-3 subtest of Grammatical Morphemes ($r = .50$) were correlated significantly with fast-mapped production, but hearing status added no significant explained variance

when added in a regression model after Grammatical Morphemes; nor (vice versa) was Grammatical Morphemes significant when entered after hearing. Thus, it was the variance shared by hearing status and Grammatical Morphemes comprehension that predicted success in production of novel words in the fast-mapping tasks. Individuals who failed the hearing screening ($n = 10$) averaged 1.10 fast-mapped words ($SD = 0.99$); individuals who passed ($n = 9$) averaged 2.00 fast-mapped words ($SD = 1.22$). When we examined the correlations between NRT and fast-mapped production in these two subgroups, neither correlation was significant.

In the TD group, NRT ($n = 18$) was significantly related to fast-mapped production ($r = .44$, $p < .05$, one tailed). K-ABC could be evaluated validly only for 15 participants; it correlated significantly as well ($r = .57$). Comprehension tests (PPVT-3, $r = .45$; TACL-3 Subtest III, $r = .48$) and visual short-term memory (Bead Memory, $r = .51$) were also predictors, although none of these measures contributed significant explained variance when entered second after the NRT. Thus, for the TD group, either the NRT or K-ABC predicted, as an index of auditory short-term memory, total fast-mapped production, as we anticipated; when we inserted both variables into the equation, the resulting model was marginally significant, $F(2, 12) = 3.13$, $p < .08$, accounting for 34% of variance, with 17% of that variance shared, 15% uniquely associated with K-ABC, and 2% uniquely associated with NRT. In the DS group, however, the variance shared by hearing status and Grammatical Morpheme comprehension predicted fast-mapped production.

Discussion

Do Adolescents With DS Show Deficits in Fast-Mapping Skill?

Evidence of a deficit in fast-mapping skills in individuals with DS has been inconsistent. When only one item was presented in a hiding task (Chapman et al., 1990), fast mapping in comprehension and production did not differ from nonverbal, mental-age-matched control participants. When multiple items were presented in more complex phonological forms and embedded in phrases rather than sentence final, fast-mapped production showed deficits compared with nonverbal, mental-age-matched control participants. In our study, participants were matched on syntax comprehension level rather than nonverbal cognition. The task, novel word form, and carrier phrase were simple, and additional support to auditory memory and elicited production was provided in some conditions. Comprehension performance did not differ significantly from syntax-comprehension-matched control partici-

pants, despite the experimental control participants' requiring memory of word form rather than which object had received a novel name.

Production performance showed a trend for a group effect, but it was in the opposite direction of expectations: Adolescents with DS tended to perform better than the TD control participants in correctness of production. This group of individuals with DS differed from participants in our prior studies in being, on average, more skilled in expressive language (individuals with mean lengths of utterance [MLUs] in narratives shorter than 2.0 were excluded from this study because of the demands of the narrative tasks); however, control participants were also excluded on these grounds. The one difference emerging in the latency data indicated longer latencies in recall of novel words in the DS than the TD group but no difference in recall of common word labels. The latter finding is consistent with Marcell et al.'s (1998) study of confrontation naming. This combination of results suggests a subtle deficit in fast mapping of production, leading to longer lexical access times when vocabulary is new, but not familiar, for individuals with DS.

Relation to the Baddeley model. To the extent that AVSTM deficits might play a role in fast mapping by participants with DS, we had anticipated that the gap between fast-mapped comprehension and production would be greater for the DS than the TD group; this was not the case. We had anticipated that high memory support might improve phonological store more for individuals with DS than the TD group and, hence, show a greater effect on fast-mapped comprehension for the DS group; although group and memory support showed a significant interaction for comprehension, it was the TD rather than the DS group who benefited from memory support in correct recall. The DS participants did, however, benefit more than the TD participants in speed of correct response. We had anticipated that—to the extent that the phonological loop was deficient for the participants with DS—elicited production would benefit them more than the TD group in production. There was no interaction, however, between group and elicited production in fast-mapped production. Instead, all participants benefited from elicited production in the low-memory condition.

Does AVSTM Predict Fast Mapping?

The group with DS showed significant difficulty, compared with the control group, in successful repetition of the novel words on the NRT, on both short (one-syllable) words like those used in the fast-mapping tasks and the longest four-syllable sequences. The scoring for these tasks did not penalize responses that distorted sounds (the typical articulatory problem of

individuals with DS; see Chapman & Hesketh's, 2000, review; Kumin, 2001); only omissions or substitutions were scored incorrect. Thus, performance on the NRT partially reflected the articulatory problems of the group of individuals with DS and any problems with AVSTM for the novel forms. The NRT failed to predict variance in fast-mapped production for individuals within the DS group, however, despite its significant relation to fast-mapped production for individuals in the TD comparison group. In addition, K-ABC and the Digit Span task failed to be predictors in the DS group. Thus, our anticipation that AVSTM would vary with fast-mapped production in the DS group was not supported by our results; instead, hearing status (and its variance shared with comprehension of grammatical morphemes) predicted outcome. Individuals with DS who had hearing in the normal range fast mapped in production almost double the number of words of those who had some degree of mild hearing loss.

Fast mapping in comprehension, in contrast, was predicted by comprehension test performance, as expected: vocabulary in the case of individuals with DS and vocabulary and syntax in the case of TD individuals. Our prediction that comprehension measures would predict fast-mapping comprehension was supported.

Limitations of This Study

These findings are constrained by the number of participants who we were able to recruit and the number of novel objects and words we were able to introduce within the protocol's time limits. Although the study evaluated both memory support and elicited production, the number of elicited productions required was less than half the number of modeled examples, to reduce task demands on participants. Thus, the study cannot speak to the comparison of the two procedures but only to their individual effectiveness in the two contexts provided by the other treatment.

Implications for Intervention

Repetitions of the novel word by the examiner, in the high-memory-support condition, helped TD children but not the group with DS in fast mapping in comprehension. That is, frequency of word input affected probability of linking a novel word form and a novel object in the TD group. In production, memory support helped all participants in the no-elicited-production condition. Elicited production did not affect comprehension performance and provided some intermediate help in production for all participants. Predictors of production performance in the DS group were vocabulary and hearing status rather than ASTM measures; in particular, those who passed the hearing screening

benefited from memory support in production, and those who showed a mild loss did not.

Thus, these findings are consistent with the view that fast mapping of new noun vocabulary by individuals with DS can be improved in production through increased frequency of presentation; however, even a mild hearing loss attenuates the benefit, and, therefore, measures to improve hearing in the learning context are needed for many individuals. We adapted our presentation procedures to provide input to the child's better ear in quiet conditions with the speaker next to the child; it is clear that more involved procedures are necessary for those adolescents with some hearing loss. It may also be the case that individuals with some hearing loss will benefit from visual versions of input (e.g., signed, written) to a greater extent than spoken input.

Fast mapping of novel words in production was improved for all participants, both by more frequent encounters with the novel word and by requests to repeat the novel word, although the latter may have distracted from remembering contextually associated information, such as where the object was hidden. Thus, opportunities to use newly encountered vocabulary are also important to provide if later ability to label the novel objects is wanted.

Implications for Research

The differing effects of input and elicited production on fast mapping in comprehension and production and the differing predictors of individual differences in performance in the two domains are consistent with the Child Talk (Chapman et al., 1992) model's view of word learning as a problem of multiple sensorimotor mappings. The quick, incidental learning of words offers a useful laboratory in which to study these multiple mappings, before overlearning binds the differing domains together.

The finding that comprehension of fast-mapped words was slower for individuals with DS compared with TD individuals but that high-frequency, common-word recall was not implicates speed of lexical comprehension as a potential factor in phonological store and its feed-forward to early motor representations. Eventual overlearning of common vocabulary and more rapid comprehension might facilitate the gains in MLU shown in adolescence, although our data suggest that production naming of common objects is slower than it would be in a TD group. We did not have sufficient numbers of correct productions to evaluate group differences in latencies for novel words. By this speculative argument, however, whatever facilitates lexical access in comprehension or production would differentially improve MLU measures of DS individuals. Elsewhere, we report (Miles, Sindberg, Bridge, & Chapman, in press), for these same participants,

that picture support for narratives differentially increases MLU for DS, but not TD, groups. If this interpretation is correct, more rapid lexical access in either comprehension or production should increase MLU, however it is achieved: through rehearsal, picture support, previewing of vocabulary in lessons, priming of vocabulary in conversation, or increasing the frequency of encounter in comprehension tasks.

Our previous discussion of the MLU deficit (Chapman et al., 2002) has suggested, within Baddeley's (1992) working-memory framework, that an automatic rehearsal link between comprehension and production might be impaired in MLU deficits. The slower comprehension times for just-learned words would be consistent with the reduced or slower function of such a mechanism, although speed of lexical access in comprehension and production could also be viewed as independent mechanisms.

Following Baddeley's (2000) model, we had anticipated that memory support would affect comprehension through phonological store (with predictors from comprehension tasks) and that elicited production would affect production through the articulatory loop (with predictors from auditory-verbal short-term recall tasks). In the TD group, this was the case. In the DS group, hearing impairment significantly affected comprehension, presumably through its effect on phonological store, and the expected benefits of the experimental interventions were reduced. The auditory-verbal short-term recall measures also did not predict the DS group's fast-mapping performances.

Within the Baddeley (2000; Baddeley, Gathercole, & Papagno, 1998) model, the addition of episodic memory provides a place for long-term knowledge to be recruited at differing speeds. Research indicating whether access time, as indicated by naming speeds and by comprehension speeds, affects MLU in utterances that incorporate the same vocabulary would be a next step in evaluating the role of slower lexical access in DS syntactic deficits and its amelioration in ongoing MLU increase in adolescence.

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